

Simone Raponi

ARTIFICIAL INTELLIGENCE - MACHINE LEARNING - CYBER SECURITY

Rome, Italy

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Experience

Senior Technical Team Leader – Senior Technical Project Manager

Rome (Hybrid)

WSENSE

Nov. 2025 – Present

- Lead two cross-functional teams (AI/Computer Vision and Cybersecurity), driving R&D across 7+ subsea AI initiatives: vessel classification via hydrophones, near-real-time fish detection/tracking/classification, biomass estimation, automated pellet counting, low-bandwidth acoustic video streaming, subsea digital twins through 3/4D reconstruction, and an AI chatbot for interpreting complex telemetry data.
- Own network and device security as Designated Officer for Classified Data and Information Security, leading active monitoring and real-time mitigation of emerging cyber threats.
- Serve as Senior Technical Project Manager for NATO Restricted defense contracts, leading multi-million European and national programs: ASTERION, SEACURE (European Defence Funds), SEAFENCE (National Military Research Plan), and DEEP (Fincantieri Integrated Underwater System).

Tech Leader AI/ML Cybersecurity

Remote

EQUILY

Mar. 2023 – Nov. 2025

- Founded and led the AI Division as the 4th employee, building and managing a multidisciplinary team of ML/AI Engineers, Scientists, and Researchers; played a pivotal role in the company's growth, culminating in a €10M Series A round, and opened new market segments by enabling enterprise clients to secure previously unaddressed API perimeters.
- Personally designed and implemented core, high-impact components of Equily's engine, including the full endpoint-exploration logic (orchestrator and all exploration strategies), advanced parameter-resolution strategies, inter-parameter dependency computation, and the internal data structures for high-performance scan processing.
- Reduced API infrastructure scan times by over 70% vs. state-of-the-art and open-source solutions, and improved vulnerability-detection accuracy by over 80% vs. market competitors.
- Researched, designed, and deployed a suite of AI-driven microservices (Golang, Python) to optimize continuous API security testing, combining classical ML (supervised/unsupervised) with Generative AI (LLMs) for adaptive analysis and autonomous decision-making.

Machine Learning Scientist

La Spezia, Italy

NATO-STO CENTER FOR MARITIME RESEARCH AND EXPERIMENTATION (CMRE)

Jul. 2021 - Mar. 2023

- Drove the ML workstream of the PROMENADE European project, advancing maritime safety and surveillance technologies.
- Designed Encoder-Decoder Seq2Seq models with LSTMs for forecasting vessel trajectories using AIS and raw Radar data, reducing the error to 2.5 Nautical Miles for 9-hour long predictions—70% lower than state-of-the-art solutions.

Machine Learning Scientist Intern in the Smart Grid Lab

Doha, Qatar

IBERDROLA INNOVATION MIDDLE EAST

(Jun. - Aug. 2019) (Mar - Apr. 2021)

- Designed and deployed LSTM-based noise prediction models, optimizing noise management in Power Line Communication (PLC) systems, leading to a 35% improvement in signal integrity and a 25% boost in overall communication efficiency.
- Developed robust distributed algorithms for data acquisition across geographically dispersed PLC devices, increasing system reliability by 40% and enabling real-time, scalable monitoring with a 30% reduction in data latency.

Machine Learning and Big Data Scientist Intern

Barcelona, Spain

TELEFONICA RESEARCH

Jul. 2018 - Aug. 2018

- Conducted in-depth analysis on AI's impact on user privacy, leading to strategies that safeguarded data while preserving service quality.
- Reduced data exposure risks by 25% and cut privacy violations by 30%.

Teaching Assistant

Doha, Qatar

HAMAD BIN KHALIFA UNIVERSITY

Jan. 2018 - Apr. 2021

- Instructor of graduate and private courses, including Security Risk Analysis, Multimedia Security, Distributed Systems Security, and Open Source Intelligence in Cyberspace, equipping students with critical skills in cybersecurity and risk management.

Education

Hamad Bin Khalifa University

Doha, Qatar

PH.D. IN COMPUTER SCIENCE AND ENGINEERING (FOCUS ON AI AND CYBERSECURITY)

Jan. 2018 - Apr. 2021

- Outcomes: 1 book, 23 international papers, 2 patents, 1 encyclopedia entry. H-Index: 16, citations: 1615, as of May 2026.
- Best Ph.D. in Computer Science and Engineering Award.
- Computer Science and Engineering Outstanding Performance Award.
- Thesis: AI-driven Detection of Cybersecurity-related Patterns.

Sapienza University of Rome

MASTER'S DEGREE IN COMPUTER SCIENCE

Rome, Italy

Jan. 2016 - Oct. 2017

- Thesis: Geographical De-anonymization of Crowds in the Dark Web, Final Mark: 110/110 summa cum laude.
- Dissertation on IEEE International Conference on Distributed Computing Systems, extension on IEEE Transactions on Services Computing.

Sapienza University of Rome

BACHELOR'S DEGREE IN COMPUTER SCIENCE

Rome, Italy

Sep. 2012 - Jan. 2016

- Thesis: An Administrative Model for Attribute-Based Access Control.
- Final Mark: 110/110 summa cum laude.

Publications

PATENTS

US Patent App. 16/657,088, Methods and Systems for Verifying the Authenticity of a Remote Service

US Patent App. 16/880,525, Online Account Access Recovery System and Method Utilizing Secret Splitting

BOOKS

2021 **New Dimensions of Information Warfare**, R. Di Pietro, S. Raponi, M. Caprolu, S. Cresci, Springer Nature.

SELECTED PAPERS

2022 **PAST-AI: Physical-layer Authentication of Satellite Transmitters via Deep Learning**, G. Oligeri, S. Sciancalepore, S. Raponi, R. Di Pietro, IEEE Transactions on Information Forensics and Security (TIFS).

2022 **Sound of Guns: Digital Forensics of Gun Audio Samples meets Artificial Intelligence**, S. Raponi, I. Ali, G. Oligeri, Springer Nature Multimedia Tools and Applications.

2021 **Cryptomining Makes Noise: Detecting Cryptojacking via Machine Learning**, M. Caprolu, S. Raponi, G. Oligeri, R. Di Pietro, Computer Communications (COMCOM).

2019 **Nationality and Geolocation-Based Profiling in the Dark (Web)**, M. La Morgia, A. Mei, E. Nemmi, S. Raponi, J. Stefa, IEEE Transactions on Services Computing (TSC).

2020 **BrokenStrokes: On the (in)Security of Wireless Keyboards**, G. Oligeri, S. Sciancalepore, S. Raponi, R. Di Pietro, ACM Conference on Security and Privacy in Wireless and Mobile Networks (WISEC).

2018 **Time-Zone Geolocation of Crowds in the Dark Web**, M. La Morgia, A. Mei, S. Raponi, J. Stefa, IEEE International Conference on Distributed Computing Systems (ICDCS).

Extracurricular Activities

The Illustrated Machine Learning Website (650 Stars and 69 forks as of May 2026)

Remote

CO-FOUNDER AND TECHNICAL CONTRIBUTOR

Jan. 2023 – Present

- <https://illustrated-machine-learning.github.io/>
- Creation of visual tools to demystify the elegance and intricate workings of machine learning concepts.

International Conferences and Journals

PROGRAM COMMITTEE MEMBER

- Served as a PC member for 20+ prestigious international conferences, contributing to key decisions on research quality and conference themes.

TECHNICAL REVIEWER

- Reviewer of 50+ leading international conferences and journals, ensuring high standards of research quality and integrity.
- Technical Reviewer of the book: 10 Machine Learning Blueprints you should know for Cybersecurity.

RedHotCyber

Remote

TECHNICAL WRITER – ARTIFICIAL INTELLIGENCE TEAM

Jul. 2023 – Sep. 2024

- Author of 10+ insightful articles exploring the synergies between Cybersecurity, Artificial Intelligence, and their practical applications.

International Contests

Doha, Qatar

CO-FOUNDER AND EXECUTIVE ASSISTANT

2019

- Co-founder and organizer of the the Qatar International Cyber Security Contest, fostering global collaboration and innovation in cybersecurity.

Funded Grants

Doha, Qatar

PRINCIPAL INVESTIGATOR

- Project “*Securing Online Transactions*” funded within the Technology and Development Fund (TDF) by QNRF (\$ 90,627.90)
- Project “*Detecting Cryptojacking in Corporate Network*” funded within the Innovation Center’s Idea Development Fund Grant (\$ 13,700)
- Project “*Strong Server Authentication for Online Transactions*” funded within the Innovation Center’s Idea Development Fund Grant (\$ 13,700)
- Project “*Extending Blockchain Technology – A Novel Paradigm and Its Applications to Cybersecurity and Fintech*” funded within NPRP11S-0109-180242, by QNRF (\$ 600,700)